HITEC University Taxila

Department of Cyber Security

Course: Essentials of Artificial Intelligence

Lab Manual 2

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LAB TASK

Task 1. Count the number of strings where the string length is 2 or more and the firstand last character are same from a given list of strings. Sample List: ['abc', 'xyz', 'aba', '1221', 'xyzzyx', 'aa', '122’] Expected Result: 4

Input

lab2 Task 1.py > ...

```python
strings = ['abc', 'xyz', 'aba', '1221', 'xyzzyx', 'aa', '122']

count = 0
for s in strings:
    if len(s) >= 2 and s[0] == s[-1]:
        count += 1

print(count)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local
s:/AI Lab/lab2 Task 1.py"
4
```

Task 2. Find the sum and product of numbers in a tuple. Sample List: (1, 2, 4, 2, 6), Sum is: 15, Product is: 96

Input

lab2 Task 2.py > ...

```python
numbers = (1, 2, 4, 2, 6)

sum_val = 0
product = 1

for n in numbers:
    sum_val += n
    product *= n

print("Sum is:", sum_val)
print("Product is:", product)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Progr
s:/AI Lab/lab2 Task 2.py"
Sum is: 15
Product is: 96
```

Task 3. Write a program to print largest number in a List Sample List: (1, 2, 4, 2, 6)

Input

lab2 Task 3.py > ...

```python
numbers = [1, 2, 4, 2, 6]

largest = numbers[0]

for n in numbers:
    if n > largest:
        largest = n

print("Largest number is:", largest)
```

Output

```
s:/AI Lab/lab2 Task 3.py"
Largest number is: 6
```

Task 4. Write a Python program to remove an empty tuple(s) from a list of tuples. Sample L = [(), (), ('',), ('a', 'b'), (), ('a', 'b', 'c'), (), ('d')], Result = [('',), ('a', 'b'), ('a', 'b', 'c'), ('d')]

Input

lab2 Task 4.py > ...

```python
L = [(), (), ('',), ('a', 'b'), (), ('a', 'b', 'c'), (), ('d')]

result = []

for t in L:
    if t != ():
        result.append(t)

print("Result =", result)
```

Output

```
s:/AI Lab/lab2 Task 4.py"
Result = [('',), ('a', 'b'), ('a', 'b', 'c'), 'd']
PS S:\AI Lab>
```

Task 5. Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (‘x’, x*x). Sample: Input a number 4 Output: {'1': 1, '2': 4, '3': 9, '4': 16}

Input

lab2 Task 5.py > ...

```python
n = int(input("Input a number: "))

d = {}

for x in range(1, n+1):
    d[str(x)] = x * x

print(d)
```

Output

```
s:/AI Lab/lab2 Task 5.py"
Input a number: 4
{'1': 1, '2': 4, '3': 9, '4': 16}
PS S:\AI Lab>
```

Task 6. Create a program using Dictionary that takes Numbers as input and print it in words form Sample: Input: 1234, Output: One Two Three Four

Input

```
lab2 Task 6.py > ...
num = input("Enter a number: ")

digits = {
    '0': 'Zero',
    '1': 'One',
    '2': 'Two',
    '3': 'Three',
    '4': 'Four',
    '5': 'Five',
    '6': 'Six',
    '7': 'Seven',
    '8': 'Eight',
    '9': 'Nine'
}

for d in num:
    print(digits[d], end=" ")
```

Output

```
s:/AI Lab/lab2 Task 6.py"
Enter a number: 0123456789
Zero One Two Three Four Five Six Seven Eight Nine
PS S:\AI Lab>
```

Conclusion:

This lab reinforced core Python skills through practical exercises in strings, tuples, lists, and dictionaries. The tasks highlighted how to manipulate data, perform mathematical operations, clean datasets, and generate structured outputs. Overall, the work improved logical thinking, coding efficiency, and problem-solving ability—key foundations for professional programming and cybersecurity applications.